ModelIt!

Student Activity Pack

How Do Maps Help Us Find Things?

Mapping the Land, Water, and Features of Earth

Grade 2 | NGSS 2-ESS2-2 | Patterns

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 Have you ever used a map to find something? What did the map look like?

2 What do you think a river looks like on a map versus in real life?

3 Draw a map of your classroom — what would it look like from above?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Land Height | Water Location | Map Features

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Mountain Area

Land Height set to tall mountain

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Flat Plains

Land Height set to flat

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Valley With River

Land Height set to low valley

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

ModelIt!

Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Map Scale — How big or small the map is compared to the real world — a map of your school fits on paper, but the real school is much bigger

Map Symbols — Special shapes and colors on maps that stand for real things — like blue for water, triangles for mountains, and green for forests

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

How do maps help us understand what the land and water look like in a place we have never visited? Use evidence from your simulation to support your answers.

1 How is looking at a map different from standing on the ground looking around?

2 Why do you think water always ends up in the lowest parts of the land?

8 STEM Challenge: Build a 3D Map of Our Area

ModelIt!

YOUR ENGINEERING MISSION

Create a 3D model map that shows the land and water features of a real or imaginary area.

THE SCENARIO

A park ranger needs a 3D map to help visitors understand the land around a new nature park. Your team will build a model that shows mountains, valleys, rivers, and lakes so visitors can see what the park looks like before they explore it.

Engineering Design Thinking

1. WHAT MATERIALS WILL YOU USE TO BUILD THE LAND — CLAY, PLAYDOUGH, OR CRUMPLED PAPER?

2. HOW WILL YOU SHOW WATER — BLUE PAINT, FOIL, OR PLASTIC WRAP?

3. HOW WILL YOU LABEL THE DIFFERENT FEATURES SO VISITORS KNOW WHAT THEY ARE LOOKING AT?

4. HOW DOES YOUR 3D MAP COMPARE TO A FLAT MAP OF THE SAME AREA?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Cartographers make maps that help people explore, build roads, and understand the Earth. They use computers, satellites, and math to create detailed maps. They earn $45,000–$80,000/year.